

# Face Tracking Robot

**Project Title:** Face tracking robot using OpenCV

**Team Members:** Shijil Shijo, Karthik N Nair, Irfan S, Saheel Siraj

**Guide:** Ms. Aliya Shahul, Assistant Professor

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## Abstract:

This project is capable of detecting a human face and automatically adjusting its camera position to keep the face in the center of the frame. The system utilizes Computer Vision (OpenCV) for detection and an Arduino-based servo mechanism for movement. This project has potential applications in automated surveillance, industrial safety monitoring.

## Methodology:

- **Image Acquisition:** The camera mounted on the robot captures live video footage.
- **Processing (Laptop):** The video feed is processed in real-time using OpenCV (running on PyCharm). The algorithm detects the face and calculates its center ( $x$ ,  $y$ ) coordinates.
- **Coordinate Calculation:** The system calculates the error (difference) between the face's current position and the center of the frame.
- **Data Transmission:** These coordinates or error values are sent to the Arduino board via serial communication.
- **Actuation:** The Arduino processes the data and drives the two servo motors (Pan and Tilt) to move the camera, ensuring the face stays locked in the center.

## Applications:

- Autonomous underwater vehicle (AUVs) for gate and object detection.
- Monitoring laborers in hazardous zones to ensure safety gear compliance or track movement.
- Social robotics (companion and educational robots)
- Auto-tracking gimbals for solo content creators or desktop cameras for video conferencing.
- Obstacle avoidance robots.

## Achievements:

- **1st Prize at** KMEA Engineering College Expo 2025.
- Exhibited at MBITS College Kothamangalam Expo in both 2025 and 2026.
- Best Committed Project Award (second semester).
- Have the chance to demonstrating and teaching school students the working of our creation.

